

More SQL

MBUA 512

Class 4 – More SQL

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

- SELECT calculations - and DISTINCT
- WHERE conditions
- ORDER BY (DESC)
- GROUP BY, and aggregates (min, max, avg, count, sum)
- LIMIT n

How to write SQL queries & tips

Other things(?): Transactions?

Based on slides by Professor Markus Luczak-Roesch.

Recap: SQL commands

- `CREATE TABLE` and `ALTER TABLE` — define the structure of a table (its *schema*)
 - `INSERT INTO` , `DELETE FROM` , `UPDATE` — change the data in the table
 - `SELECT` — query data
- (there are a lot more than these)

Important: make sure you select `MySQL` (not `Trino`) and your own workspace
(`user_[username]`)

SELECT calculations

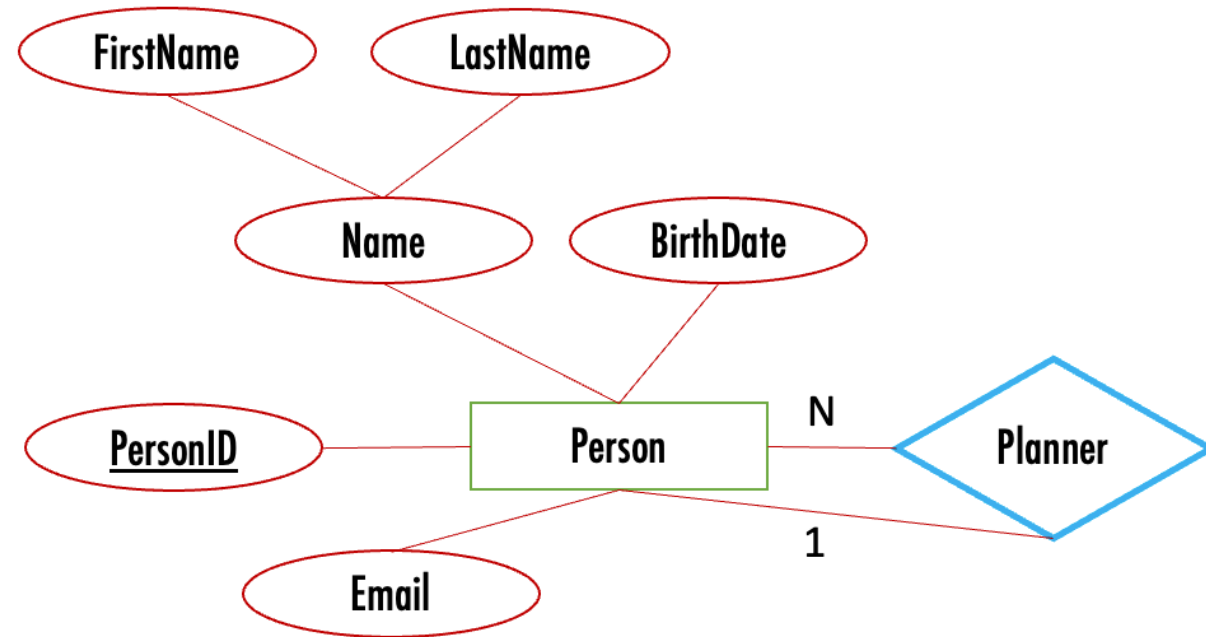
- string ||
- AS
- numerical operations including floor, ceiling, round
- date operations - including current date, subtraction

WHERE conditions

- IS NULL
- IN
- LIKE and %_
- numerical conditions
- date conditions
- logical operators (AND, OR, NOT)
- (not) EXISTS
- UNION, INTERSECT

Running Example: birthday

- Scenario: send a message to people on their birthday, and remind their designated party planner to plan a party earlier
- Data: name (first, last), birthdate, email address, and designated party planner



To create the database

```
DROP TABLE IF EXISTS people;

CREATE TABLE people (
    personID INTEGER PRIMARY KEY AUTOINCREMENT,
    firstName VARCHAR(30) NOT NULL,
    lastName VARCHAR(30),
    birthdate DATE NOT NULL,
    planner INTEGER,
    email VARCHAR(30),
    FOREIGN KEY(planner) REFERENCES people(personID)
);
```

To add data

```
INSERT INTO people VALUES
    (1, 'Bob', 'Smith', '1917-01-03', 1, NULL),
    (2, 'Bob', 'Smith', '1973-03-25', NULL, 'bob@smith.net'),
    (3, 'Alice', 'Smith', '2026-07-07', 1, 'Alice@smith.net'),
    (4, 'sam', 'Smith', '2020-01-03', 1, 'sam'),
    (5, 'Bobbie', 'Smithton', '1987-03-16', NULL, 'bob@smith.net');
```


Some queries to write (1/2)

1. Find people whose birthday is today so we can email them (combine first and last names)
2. Find people with a special birthday (e.g. 100) coming up (hint: need to calculate age, use FLOOR)
3. Find people with a birthday in January, February or March so we can email their planner (hint: use IN)

Some queries to write (2/2)

administrative tasks:

4. Find people with missing or invalid email addresses (hint: use NULL AND ... NOT LIKE)
5. Find where two (or more) people have the same email address (hint: use EXISTS)
6. Find people who are not planners for anyone else and do not have a birthday (hint: use EXISTS)

analysis task:

7. how many people have a birthday in each month? (hint: use GROUP BY and EXTRACT(MONTH from ...))

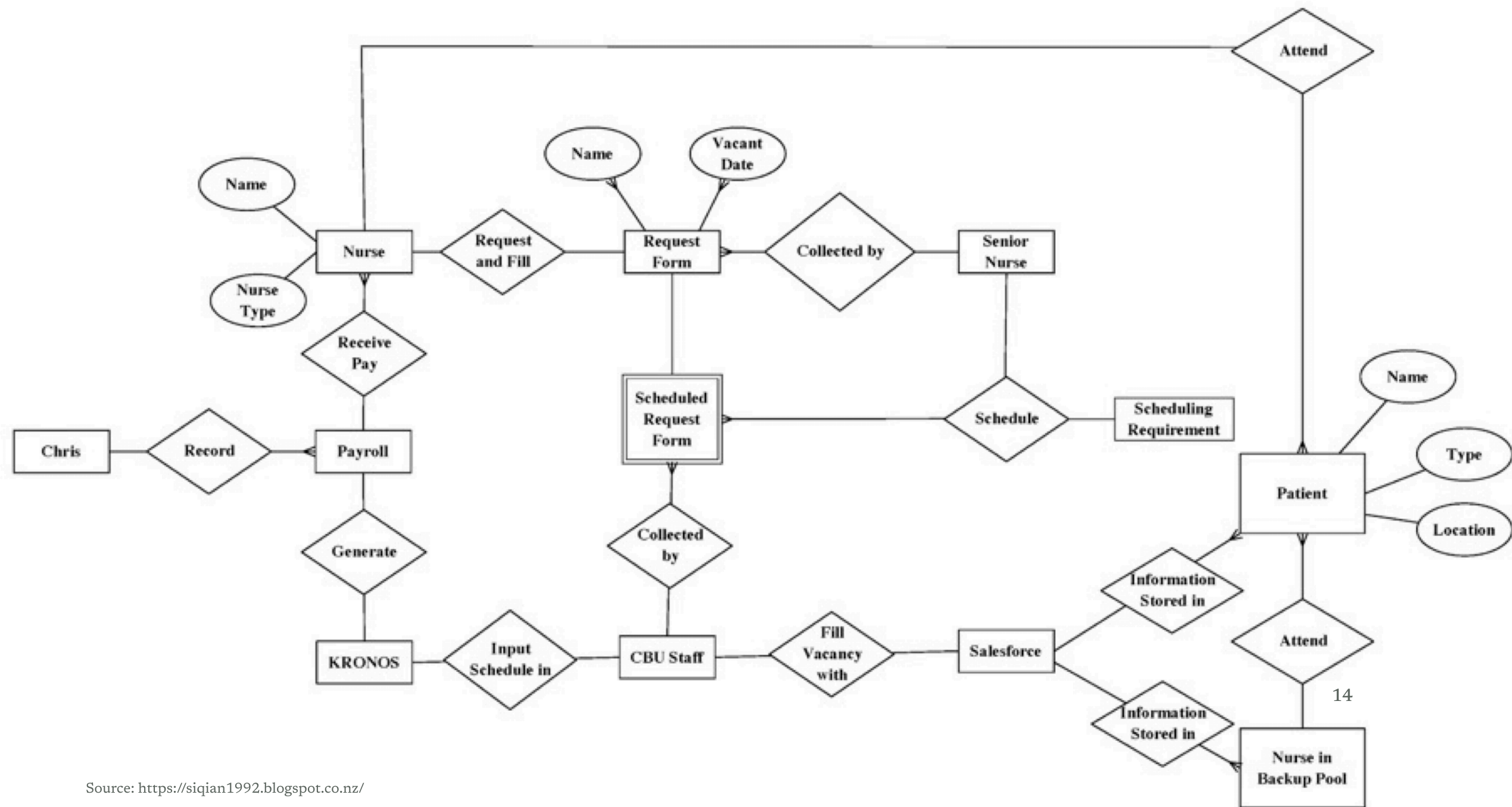
Tips for writing SQL

- Build it up in pieces, testing each piece (* can be useful)
- Think about:
 - What information you need to see in the result (which columns ...) ... values?
Calculations? Aggregate calculations (e.g. SUM)?
 - What tables they come from
 - How you need to JOIN the tables
 - What constraints to use ... and how to express them
 - WHERE if based on row
 - WHERE ... AND exists – if not

More tips

- `' '` (single quotes), not `" "` (double quote)
- Distinct – the whole row needs to be distinct, so sometimes select fewer columns
- Group By ... Having ... - the "Having" goes just after the "Group By ..."
- NULL is not the same as the empty string
- Too many rows in result? Check JOIN conditions (or using "NATURAL JOIN" - which is not the same as "JOIN")
- No rows in result? Check conditions, and perhaps remove some
- Using exists – make sure there is also an "AS ..." (so can refer to parent query in sub-query)

Summary



Activities

Activities – More SQL

1. Go to superset.runs.nz and login if needed
2. Select MySQL (not Trino) and your own workspace (user_[username])
3. If needed, create the flight DB using [this file](#)

Write queries to answer the following questions

TO COMPLETE

Activity – ...

~ 10 minutes, in groups